Parag Mollick

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Education

Bangladesh University of Engineering and Technology (BUET) 2022 – Present
Bachelor of Science in Mechanical Engineering

Cantonment College Jashore 2019 – 2021
Higher Secondary Certificate (HSC) – Science, GPA-5.00/5.00

Satkhira Government High School 2019
Secondary School Certificate (SSC) – Science, GPA-5.00/5.00

Teaching Experience

Private Tutor 2023 – Present
Mathematics and Physics (Higher Secondary and admission Level)

- Provided private tuition to HSC-level students for 3 years, focusing on conceptual clarity in Mathematics and Physics

Answer Script Evaluator 2022 – 2023
Udvash-Unmesh Academic and Admission Care

- Evaluated answer scripts of university admission test preparation exams, ensuring accurate and consistent grading

Research Experience

Undergraduate Thesis June 2026 – Present
Department of Mechanical Engineering, BUET

- "Numerical Investigation of Fin Geometry Effects on a Rear Door Heat Exchanger (RDHx) Using Ternary Hybrid Nanofluid for AI Server Rack Cooling" *(in progress)*
- Conducted an extensive literature review on hybrid/ternary nanofluid-based AI server thermal management systems , identifying key research gaps in geometry-nanofluid-magnetic field coupling
- Building simulation proficiency in COMSOL Multiphysics (CFD, Heat Transfer, and AC/DC modules) for coupled fluid-thermal-magnetic field analysis

Publications

Under Review / Submitted

- Parag Mollick et al., "Design, Experimental Validation, and Numerical Coolant Comparison of a Compact Plate-Fin-and-Tube Rear Door Heat Exchanger with Al_2O_3 Nanofluid for Data Center Cooling," submitted to *ICMIME 2026* (Thermal Engineering Track), under review

Projects

AeroMark (ME 366 Project)
FPV Drone-Based Wildlife Tranquilization System

- Co-developed an FPV drone-based system for remote wildlife tranquilization, integrating a custom electromagnetic coil gun and stabilized camera for research, conservation, and veterinary support
- Coil gun boosts a 12V supply to high voltage, charging capacitors that discharge through a solenoid to launch a metal dart via induced magnetic field

Radiator-Type Rear Door Heat Exchanger Equipment
Design Project, Dept. of ME, BUET

- Designed and analyzed a compact plate-fin-and-tube rear door heat exchanger with Al_2O_3 nanofluid for data center cooling applications, leading to an ICMIME 2026 conference submission

Design and Optimization of Spiral Jaw Clutch
SolidWorks

- Designed and optimized a spiral jaw clutch mechanism for improved torque transmission

Honors and Awards

- Government Board Scholarship – Higher Secondary Certificate (HSC), for exceptional academic excellence

- Government Board Scholarship – Secondary School Certificate (SSC), top merit position, Jashore Education Board
- Government Talentpool Scholarship – Junior School Certificate (JSC)
- Government Talentpool Scholarship – Primary School Certificate (PSC)
- Prize positions in Badminton – Annual Sports & Cultural Competition, Satkhira Govt. High School (2016, 2017)

Technical Skills

Computer Languages: C, Python, MATLAB
MATLAB Certifications: MATLAB Onramp, Solving Nonlinear Equations with MATLAB, Signal Processing Onramp – MathWorks Training Services (2024–2025)
Research Software & Tools: COMSOL Multiphysics (learning), ANSYS Simulation (Getting Started with Ansys Mechanical – Ansys, December 2025)
CAD Software: SolidWorks (Certified SolidWorks Associate – CSWA), AutoCAD

Co-curricular Activities

- **Interplanetar Workshop 2024** – Successfully completed the Mechanical course (July 10–18, 2024)
- **Intra BUET Robo Challenge 2024** – Participated in the Soccer Bot Challenge segment, organized by BUET Robotics Society
- **Nazrul o Lok Sangeet (Vocal Music)** – Completed a 5-year structured training program, Dhrubo Parishad Bangladesh (2010–2016), attaining the "Sangeet Visharad" title with 1st position; also secured 1st position in both categories at Satkhira Govt. High School's Annual Sports and Cultural Competition (2016–2017)
- **Charu o Karu Shilpo (Arts and Crafts)** – Completed a multi-year structured program (2008–2016), Dhrubo Parishad Bangladesh; attained final title with 2nd position
- **Poetry Recitation** – 1st Division, 1st Year, Dhrubo Parishad Bangladesh (2008–2009)

Professional Experience

Industrial Attachment Trainee June 2026
Meghna Group of Industries (MGI)

- Gained hands-on experience in large-scale industrial operations and mechanical maintenance; observed manufacturing processes and safety protocols in a leading industrial setup